Solid Geometry & Similarity

ANSWER KEY

Section 1: Polyhedra & Euler's Formula

1 A convex polyhedron has 12 vertices and 30 edges. How many faces does it have?

a. 18

b. 20 ✓

c. 22

d. 24

By Euler's formula $V - E + F = 2$: $12 - 30 + F = 2 \Rightarrow -18 + F = 2 \Rightarrow F = 20$ (an icosahedron).

2 What is the total surface area of a regular tetrahedron with edge length 6 cm?

a. $36\sqrt{3}$ **cm**^{2} ✓

b. $18\sqrt{3}$ cm^2

c. 54 cm^2

d. $72\sqrt{3}$ cm^2

A regular tetrahedron has 4 equilateral triangular faces: $4 \times \frac{\sqrt{3}}{4}(6^2) = 36\sqrt{3}$ cm^2.

3 What is the length of the space diagonal of a rectangular prism with dimensions 3, 4, and 12?

a. 13 ✓

b. 15

c. $\sqrt{160}$

d. 14

$d = \sqrt{l^2 + w^2 + h^2} = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

Section 2: Cones, Pyramids & Frustums

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4 A right pyramid has a square base of side 10 cm and an altitude of 12 cm. What is its lateral surface area?

a. 260 **cm**^{2} ✓

b. 240 cm^2

c. 360 cm^2

d. 130 cm^2

Slant height $l = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = 13$ cm. Lateral area $= \frac{1}{2}Pl = \frac{1}{2}(40)(13) = 260$ cm^2.

5 A right circular cone of radius 6 cm and height 9 cm is cut by a plane parallel to its base at height 3 cm from the base. What is the radius of the top circular cross section?

a. 4 **cm** ✓

b. 3 cm

c. 4.5 cm

d. 5 cm

Height of smaller top cone $= 9 - 3 = 6$ cm. By similarity: $\frac{r_{\text{top}}}{6} = \frac{6}{9} \Rightarrow r_{\text{top}} = 6 \times \frac{2}{3} = 4$ cm.

6 What is the volume of the frustum formed in the above cone with bases of radii 6 cm and 4 cm and height 3 cm?

a. 76π **cm**^{3} ✓

b. 72π cm^3

c. 84π cm^3

d. 96π cm^3

Volume of frustum: $\frac{1}{3}\pi h(R^2 + Rr + r^2) = \frac{1}{3}\pi(3)(36 + 24 + 16) = 76\pi$ cm^3.

Section 3: Spheres, Cylinders & Cross-Sections

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7 A cylinder is circumscribed around a sphere of radius r. What is the ratio of the volume of the sphere to the volume of the cylinder?

a. $\frac{2}{3}$ ✓

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{4}{5}$

Cylinder height $h = 2r$. Volume of cylinder $= \pi r^2(2r) = 2\pi r^3$. Volume of sphere $= \frac{4}{3}\pi r^3$. Ratio $= \frac{4/3}{2} = \frac{2}{3}$ (Archimedes' theorem).

8 A sphere is intersected by a plane 5 cm from its center. If the radius of the cross-sectional circle is 12 cm, what is the radius of the sphere?

a. 13 **cm** ✓

b. 15 cm

c. $\sqrt{119}$ cm

d. 17 cm

$R = \sqrt{d^2 + r_{\text{circle}}^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ cm.

9 What is the total surface area of a closed hemisphere of radius 4 cm?

a. 48π **cm**^{2} ✓

b. 32π cm^2

c. 64π cm^2

d. 16π cm^2

Total surface area $=$ Curved area $+$ Base area $= 2\pi r^2 + \pi r^2 = 3\pi r^2 = 3\pi(16) = 48\pi$ cm^2.

Section 4: Scale Factors & Volume Ratios

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10 Two similar solid statues have heights in the ratio $2:5$. If the smaller statue weighs 16 kg, what does the larger statue weigh (assuming identical material)?

a. 250 **kg** ✓

b. 100 kg

c. 200 kg

d. 320 kg

Volume and weight scale with the cube of the linear factor: $(\frac{5}{2})^3 = \frac{125}{8}$. Weight $= 16 \times \frac{125}{8} = 250$ kg.

11 If the volume of a sphere is multiplied by 27, by what factor does its surface area increase?

a. 9 ✓

b. 3

c. 6

d. 18

Volume factor $= 27 = k^3 \Rightarrow k = 3$. Surface area factor $= k^2 = 3^2 = 9$.

12 A solid cylinder has its radius doubled and its height halved. How does its volume change?

a. **Doubled** ✓

b. Halved

c. Quadrupled

d. Remains unchanged

Original volume $= \pi r^2 h$. New volume $= \pi(2r)^2(\frac{h}{2}) = \pi(4r^2)(\frac{h}{2}) = 2\pi r^2 h$. The volume is doubled.

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